ADVENTURES IN BIG MATH · 3RD GRADE MATH

MISSION 01 OF 08 · WORKSHEET PACK

Multiplication Masters

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Arrays Become Area

Warm-Up facts, then break each big number into tens and ones.

Warm-Up

9 ITEMS · RECALL

1 6×7 ____
2 8×4 ____
3 9×6 ____
4 7×7 ____
5 2×7 ____
6 5×6 ____
7 3×4 ____
8 2×9 ____
9 5×9 ____

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

10. 8×8 ....

★ Challenge NOT GRADED · REASON IT OUT

11. 12×3

12. 11×5

13. 4×13 (40 + 12)

14. 6×14 (60 + 24)

15. 3×27 (60 + 21)

16. 5×18 (50 + 40)

17. 7×16 (70 + 42)

18. Area 84, one side 6. Other side? (84 ÷ 6)

19. How many rectangles have area 36? (1×36, 2×18, 3×12, 4×9, 6×6)

20. 10×8

21. 12×6

22. 11×9

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal ONE SENTENCE PER MISTAKE

Breaking Apart

One-digit times two-digit. Two rooms every time.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

9 ITEMS · RECALL

1 5 × 9	2 8 × 8	3 7 × 8	4 6 × 9
5 3 × 6	6 4 × 5	7 2 × 8	8 5 × 5
9 3 × 9			

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. 4 × 9

11. 7 × 6

12. 9 × 9

★ Challenge

NOT GRADED · REASON IT OUT

13. 4×12

14. 3×11

15. 8×34 (240 + 32)

16. 6×47 (240 + 42)

17. 9×26 (180 + 54)

18. 7×58 (350 + 56)

19. 4×96 (360 + 24)

20. $6 \times 4\square = 2\blacktriangle 6$ — what is $\square$? (6 × 46 = 276)

21. $4 \times 68 = ?$ (Same as 8 × 34)

22. 10×7

23. 12×8

24. 7×12

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Mental Math Moves

Go past the number, then take back the extra. No pencil.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

1 ITEMS · RECALL

1 10 × 12

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

2. 9 × 6 — go to 10 × 6, then take one 6 back ....

3. 5 × 8 — half of 10 × 8 ....

4. 6 × 9 — go to 6 × 10, then take one 6 back ....

Challenge

NOT GRADED · REASON IT OUT

5. 25 × 4

6. 25 × 8

7. 50 × 6

8. 20 × 7

9. 15 × 4

10. 30 × 9

11. 19 × 6 (20 × 6 - 6)

12. 99 × 7 (700 - 7)

13. 48 × 5 (Half of 480)

14. 25 × 12 (100 × 3)

15. 102 × 4 (400 + 8)

16. 35 × 6 (180 + 30)

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

The Four Rooms

Two 2-digit numbers make four rooms. Count them before you add.

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Warm-Up

6 ITEMS · RECALL

1 10 × 10

2 20 × 30

3 40 × 50

4 12 × 10

5 60 × 70

6 90 × 80

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

7. 7 × 8 as two rooms of 7 × 4 — the total

8. A rectangle 6 by 9 — the area

9. Area 72, one side 8 — the other side

10. 4 × 30

11. 6 × 40

Challenge

NOT GRADED · REASON IT OUT

12. 23 × 14 (200 + 80 + 30 + 12)

13. 36 × 25 (600 + 150 + 120 + 30)

14. 47 × 32 (1200 + 210 + 80 + 14)

15. 58 × 46 (2000 + 320 + 300 + 48)

16. 64 × 27 (1200 + 420 + 80 + 28)

17. In 23 × 14, which two rooms make 92? Type their sum. (80 + 12)

18. 25 × 25 = 625. What is 26 × 26? (625 + 25 + 25 + 1)

19. 11 × 2

20. 12 × 3

21. 13 × 2

22. 20 × 4

23. 15 × 2

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Rectangle Hunt & Puzzles

Enrichment day. Reason it out — no guessing.

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Warm-Up

2 ITEMS · RECALL

1 4×4

2 6×6

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

3. 9×7

4. 8×6

★ Challenge
NOT GRADED · REASON IT OUT

5. 6 × 4☐ = 276 → ☐ (6 × 46 = 276)

6. 2☐3 × 4 = 932 → ☐ (932 ÷ 4 = 233)

7. 1☐ × 15 = 240 → ☐ (16 × 15 = 240)

8. 3☐ × 7 = 252 → ☐ (36 × 7 = 252)

9. ☐☐ × 11 = 484 → ☐☐ (484 ÷ 11 = 44)

10. Rectangles with area 48? (1×48, 2×24, 3×16, 4×12, 6×8)

11. Rectangles with area 37? (37 is prime)

12. Rectangles with area 100? (1×100, 2×50, 4×25, 5×20, 10×10)

13. 25 × 4

14. 15 × 12

15. 99 × 3

✎ Working space
SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal
ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

1.1 · Arrays Become Area

OUT OF 10

Warm-Up · 1–9 · 42 | 32 | 54 | 49 | 14 | 30 | 12 | 18 | 45

Core · 10 · 64

Challenge · 11–22 · 36 | 55 | 52 | 84 | 81 | 90 | 112 | 14 | 5 | 80 | 72 | 99

1.2 · Breaking Apart

OUT OF 12

Warm-Up · 1–9 · 45 | 64 | 56 | 54 | 18 | 20 | 16 | 25 | 27

Core · 10–12 · 36 | 42 | 81

Challenge · 13–24 · 48 | 33 | 272 | 282 | 234 | 406 | 384 | 6 | 272 | 70 | 96 | 84

1.3 · Mental Math Moves

OUT OF 4

Warm-Up · 1 · 120

Core · 2–4 · 54 | 40 | 54

Challenge · 5–16 · 100 | 200 | 300 | 140 | 60 | 270 | 114 | 693 | 240 | 300 | 408 | 210

1.4 · The Four Rooms

OUT OF 11

Warm-Up · 1–6 · 100 | 600 | 2000 | 120 | 4200 | 7200

Core · 7–11 · 56 | 54 | 9 | 120 | 240

Challenge · 12–23 · 322 | 900 | 1504 | 2668 | 1728 | 92 | 676 | 22 | 36 | 26 | 80 | 30

Fri · Rectangle Hunt & Puzzles

OUT OF 4

Warm-Up · 1–2 · 16 | 36

Core · 3–4 · 63 | 48

Challenge · 5–15 · 6 | 3 | 6 | 6 | 44 | 5 | 1 | 5 | 100 | 180 | 297

Where the Rooms Hide

Every line of the algorithm is one of the four rooms. Find it before you trust it.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 20×10

2 3×4

3 Add 200, 30, 80 and 12

4 So 23×14 is

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

5. 8×7 as $8 \times 5 + 8 \times 2$ — the total ....

6. The 8×5 room of 8×7 ....

★ Challenge

NOT GRADED · REASON IT OUT

7. 3×10

8. 20×4

9. 31×22 — the tens×tens room (30×20)

10. 31×22 — the ones×ones room

11. 31×22 altogether

12. 45×23

13. 18×26

14. In 47×35, which room is biggest — type its value (40×30)

15. 47×35

16. 2×20

17. 3×30

18. 4×20

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Fluency with Carrying

Algorithm only. Use the area model to check, not to solve.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

6 ITEMS · RECALL

1 14 × 12

2 21 × 13

3 32 × 11

4 15 × 15

5 24 × 12

6 13 × 13

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

7. 7 × 6

8. 9 × 8

9. 6 × 6

10. 5 × 70

11. 8 × 40

★ Challenge

NOT GRADED · REASON IT OUT

12. 38 × 24
13. 56 × 27
14. 49 × 36
15. 63 × 48
16. 75 × 29
17. 87 × 64
18. 99 × 99 (100 × 99 take one 99)
19. 2 × 13
20. 3 × 12
21. 4 × 11
22. 2 × 24
23. 3 × 21

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Three Digits by One

A longer rectangle, split into three rooms instead of two.

Warm-Up 7 ITEMS · RECALL

1 9 × 8	2 10 × 10	3 20 × 20	4 30 × 10
5 40 × 30	6 50 × 20	7 60 × 30	

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

8. 3 × 24 as 3 × 20 + 3 × 4 ....
9. The 3 × 20 room of 3 × 24 ....
10. 2 × 43 as 2 × 40 + 2 × 3 ....
11. 7 × 60 ....
12. 9 × 30 ....

★ Challenge

NOT GRADED · REASON IT OUT

13. 100 × 4
14. 200 × 3
15. 30 × 7
16. 400 × 2
17. 50 × 6
18. 234 × 3 (600 + 90 + 12)
19. 417 × 5
20. 326 × 4
21. 508 × 7
22. 739 × 6
23. 1,245 × 8
24. A number × 6 is 2,742. The number

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Estimate First

Round, predict, compute, compare. A wrong answer should look wrong.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

5 ITEMS · RECALL

1 Estimate 19 × 21 by rounding both

2 Estimate 29 × 31

3 Estimate 48 × 12

4 Estimate 39 × 41

5 Estimate 22 × 18

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

6. Round 48 to the nearest ten

7. Estimate 4 × 19 by rounding 19 to 20

8. 4 × 19 — the true value

9. Estimate 3 × 31 by rounding 31 to 30

10. 3 × 31 — the true value

Challenge

NOT GRADED · REASON IT OUT

11. Estimate 51 × 9
12. Estimate 187 × 4
13. True value of 187 × 4
14. Estimate 62 × 38
15. True value of 62 × 38
16. Estimate 296 × 5
17. Someone says 43 × 27 = 301. Type the real answer (They only multiplied by the 7)
18. Estimate 412 × 19
19. 5 × 12
20. 6 × 11
21. 4 × 25
22. 2 × 50

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Lattice Detour

Enrichment. A 500-year-old method — then decide whether it beats yours.

Warm-Up
4 ITEMS · RECALL

1 6 × 7

2 Digits in the answer to 23 × 14

3 7 × 5

4 9 × 4

Core
GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. Lattice cells needed for 2 digits × 2 digits

6. Lattice cells for 3 digits × 2 digits

7. Rows of 6 with 7 in each — how many

8. Area 45, one side 5 — the other side

Challenge

NOT GRADED · REASON IT OUT

9. 23 × 14 by lattice — the answer
10. 456 × 23 — the answer
11. Partial products in a 3-by-2 lattice
12. 12 × 12
13. 16 × 15
14. 35 × 24
15. 52 × 18
16. 125 × 8

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 · Where the Rooms Hide

OUT OF 6

Warm-Up · 1–4 · 200 | 12 | 322 | 322

Core · 5–6 · 56 | 40

Challenge · 7–18 · 30 | 80 | 600 | 2 | 682 | 1035 | 468 | 1200 | 1645 | 40 | 90 | 80

2.2 · Fluency with Carrying

OUT OF 11

Warm-Up · 1–6 · 168 | 273 | 352 | 225 | 288 | 169

Core · 7–11 · 42 | 72 | 36 | 350 | 320

Challenge · 12–23 · 912 | 1512 | 1764 | 3024 | 2175 | 5568 | 9801 | 26 | 36 | 44 | 48 | 63

2.3 · Three Digits by One

OUT OF 12

Warm-Up · 1–7 · 72 | 100 | 400 | 300 | 1200 | 1000 | 1800

Core · 8–12 · 72 | 60 | 86 | 420 | 270

Challenge · 13–24 · 400 | 600 | 210 | 800 | 300 | 702 | 2085 | 1304 | 3556 | 4434 | 9960 | 457

2.4 · Estimate First

OUT OF 10

Warm-Up · 1–5 · 400 | 900 | 500 | 1600 | 400

Core · 6–10 · 50 | 80 | 76 | 90 | 93

Challenge · 11–22 · 500 | 800 | 748 | 2400 | 2356 | 1500 | 1161 | 8000 | 60 | 66 | 100 | 100

Fri · Lattice Detour

OUT OF 8

Warm-Up · 1–4 · 42 | 3 | 35 | 36

Core · 5–8 · 4 | 6 | 144 | 240

Challenge · 9–14 · 322 | 10488 | 6 | 840 | 936 | 1000

Factor Pairs

Every rectangle you can build is a factor pair.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

8 ITEMS · RECALL

1 Factor pairs of 12 — how many (1×12, 2×6, 3×4)	2 Smallest factor of any number	3 12 ÷ 3	4 Is 5 a factor of 20 — yes or no
5 Is 7 a factor of 20	6 Largest factor of 20	7 6 × 8	8 7 × 9

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

9. Factor pairs of 24 — how many ....

10. Factor pairs of 36 — how many ....

11. All factors of 18 — how many ....

12. Factors of 30 — how many ....

13. The missing partner of 4 in 48 ....

★ Challenge

NOT GRADED · REASON IT OUT

14. A number under 30 with exactly 8 factors

15. Highest factor of 48 below 48

16. 4 × 12

17. 5 × 14

18. 3 × 15

19. 2 × 45

20. 18 × 6

21. 24 × 7

22. 36 × 5

23. 48 × 4

24. 27 × 8

25. 144 × 6

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Factor Rainbows

Draw the arcs. Where they stop is the question worth asking.

Warm-Up 12 ITEMS · RECALL

1 Factors of 16 — how many	2 Partner of 2 in 16	3 Partner of 1 in 16	4 Middle factor of 16
5 Factors of 9 — how many	6 Middle factor of 9	7 Rows of 4, 3 rows — total	8 Rows of 5, 4 rows
9 Rows of 6, 3 rows	10 Rows of 10, 7 rows	11 Rows of 2, 9 rows	12 Rows of 8, 2 rows

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Highest number you must test for factors of 36 (Its square root) ....
14. Highest you must test for 100 ....
15. Factors of 100 — how many ....
16. Arcs in the rainbow of 24 ....
17. Factors of 49 — how many ....
18. Rows of 12, 8 rows ....
19. Rows of 15, 6 rows ....
20. Rows of 14, 7 rows ....
21. Rows of 24, 5 rows ....
22. Rows of 16, 9 rows ....

★ Challenge

NOT GRADED · REASON IT OUT

23. A rainbow with an unpaired middle means the number is a

24. Factors of 144 — how many

25. Rows of 25, 24 rows

26. Rows of 36, 15 rows

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Common Factors

Two numbers, the factors they share, and the biggest of those.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up 12 ITEMS · RECALL

1 A factor shared by every pair of numbers	2 Common factors of 6 and 9 — the greatest	3 Greatest common factor of 4 and 8	4 GCF of 5 and 10
5 GCF of 3 and 7	6 GCF of 12 and 12	7 2 × 6	8 6 × 2
9 3 × 7	10 7 × 3	11 4 × 8	12 8 × 4

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. GCF of 12 and 18
14. GCF of 24 and 36
15. GCF of 15 and 25
16. GCF of 16 and 40
17. GCF of 27 and 45
18. (2 × 3) × 5
19. 2 × (3 × 5)
20. 5 × 7 × 2
21. 8 × 5 × 3

★ Challenge
NOT GRADED · REASON IT OUT

22. GCF of 48 and 72

23. Two numbers whose only common factor is 1 are called

24. $4 \times 25 \times 2$

25. $2 \times 3 \times 4 \times 5$

26. $25 \times 4 \times 11$

✎ Working space
SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal
ONE SENTENCE PER MISTAKE

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Square Numbers

Why do squares have an odd number of factors? Build the rectangles and see.

Warm-Up

11 ITEMS · RECALL

1 3 × 3	2 5 × 5	3 7 × 7	4 10 × 10
5 4 × 4	6 8 × 8	7 9 × 2	8 9 × 3
9 9 × 5	10 9 × 7	11 9 × 8	

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Factors of 25 — how many ....

13. Factors of 36 — how many ....

14. The square number between 60 and 70 ....

15. The square number between 40 and 50 ....

16. A square with side 6 — the area ....

Challenge

NOT GRADED · REASON IT OUT

17. 12×12

18. 15×15

19. Square numbers between 1 and 100 — how many

20. A square's factor count is always odd or even — type one

21. 9×10

22. 9×12

23. 9×15

24. 11×7

25. 11×12

26. 12×9

27. 99×7

28. 999×4

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

3 ITEMS · RECALL

1 7×8

2 8×3

3 7×4

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

4. Factor pairs of 24 — how many

5. GCF of 12 and 18

6. 8×7

7. A rectangle 7 by 9 — the area

★ Challenge

NOT GRADED · REASON IT OUT

8. 6×40
9. 8×34
10. 23×14
11. Estimate 39×21
12. Area 84, one side 6 — the other
13. 14×6
14. 23×9
15. 45×32
16. 67×21
17. 250×4

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 · Factor Pairs

OUT OF 13

Warm-Up · 1–8 · 3 | 1 | 4 | yes | no | 20 | 48 | 63

Core · 9–13 · 4 | 5 | 6 | 8 | 12

Challenge · 14–25 · 24 | 24 | 48 | 70 | 45 | 90 | 108 | 168 | 180 | 192 | 216 | 864

3.2 · Factor Rainbows

OUT OF 22

Warm-Up · 1–12 · 5 | 8 | 16 | 4 | 3 | 3 | 12 | 20 | 18 | 70 | 18 | 16

Core · 13–22 · 6 | 10 | 9 | 4 | 3 | 96 | 90 | 98 | 120 | 144

Challenge · 23–26 · square | 15 | 600 | 540

3.3 · Common Factors

OUT OF 21

Warm-Up · 1–12 · 1 | 3 | 4 | 5 | 1 | 12 | 12 | 12 | 21 | 21 | 32 | 32

Core · 13–21 · 6 | 12 | 5 | 8 | 9 | 30 | 30 | 70 | 120

Challenge · 22–26 · 24 | coprime | 200 | 120 | 1100

3.4 · Square Numbers

OUT OF 16

Warm-Up · 1–11 · 9 | 25 | 49 | 100 | 16 | 64 | 18 | 27 | 45 | 63 | 72

Core · 12–16 · 3 | 9 | 64 | 49 | 36

Challenge · 17–28 · 144 | 225 | 10 | odd | 90 | 108 | 135 | 77 | 132 | 108 | 693 | 3996

Fri · Mid-Unit Quiz

OUT OF 11

Warm-Up · 1–4 · 56 | 240 | 24 | 28

Core · 5–11 · 272 | 322 | 4 | 6 | 800 | 84 | 207

Challenge · 12–15 · 14 | 1440 | 1407 | 1000

NAME __________

DATE __________

RIGHT ____ OF ____ TRIED

Multiples

Skip-count on the 100-grid and watch where the patterns cross.

Warm-Up

9 ITEMS · RECALL

1 Third multiple of 4

2 Fifth multiple of 3

3 Fourth multiple of 6

4 Is 21 a multiple of 3 — yes or no

5 Is 22 a multiple of 3

6 3 × 3

7 5 × 8

8 6 × 4

9 4 × 7

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. First number that is a multiple of both 4 and 6
11. First multiple of both 3 and 5
12. First multiple of both 6 and 8
13. Multiples of 7 below 50 — how many
14. First multiple of both 4 and 10

★ Challenge

NOT GRADED · REASON IT OUT

15. Tenth multiple of 5
16. Lowest common multiple of 9 and 12
17. Numbers below 100 that are multiples of both 3 and 4
18. 2 × 12
19. 10 × 9
20. 13 × 7
21. 17 × 6
22. 19 × 8
23. 22 × 9
24. 31 × 7
25. 123 × 8
26. 307 × 6

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Divisibility Rules

Tests for 2, 3, 5, 9 and 10 — and why the 3-rule works at all.

Warm-Up

12 ITEMS · RECALL

1 Is 48 divisible by 2 — yes or no	2 Is 35 divisible by 5	3 Is 47 divisible by 2	4 Is 90 divisible by 10
5 Digit sum of 51	6 Is 51 divisible by 3	7 $2 each, 4 items	8 $3 each, 5 items
9 $5 each, 6 items	10 $10 each, 3 items	11 $4 each, 4 items	12 $6 each, 2 items

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Digit sum of 738
14. Is 738 divisible by 9 — yes or no
15. Is 4,671 divisible by 3
16. Is 1,234 divisible by 3
17. Smallest digit that makes 12_ divisible by 3
18. $12 each, 7 items
19. $15 each, 6 items
20. $24 each, 5 items
21. $18 each, 9 items
22. $35 each, 4 items

★ Challenge

NOT GRADED · REASON IT OUT

23. A number divisible by both 2 and 3 is divisible by

24. Is 9,999 divisible by 9 — yes or no

25. $125 each, 8 items

26. $47 each, 12 items

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Primes & Composites

Colour the sieve. Every prime below 100 on one page.

Warm-Up

6 ITEMS · RECALL

1 Is 7 prime — yes or no

2 Is 9 prime

3 Is 2 prime

4 Is 1 prime

5 Factors a prime has

6 The only even prime

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

7. Primes below 20 — how many
8. Primes below 50 — how many
9. The prime just after 23
10. Is 51 prime — yes or no (3 × 17)
11. The largest prime below 100
12. 12 × 11
13. 14 × 11
14. 23 × 11
15. 35 × 11
16. 42 × 11

★ Challenge

NOT GRADED · REASON IT OUT

17. Primes below 100 — how many
18. Is 91 prime — yes or no (7 × 13)
19. 2 × 11
20. 3 × 11
21. 4 × 11
22. 5 × 11
23. 6 × 11
24. 7 × 11
25. 87 × 11
26. 123 × 11

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Prime Factor Trees

Break a number all the way down until only primes are left.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up 12 ITEMS · RECALL

1 12 as 2 × ?	2 6 as 2 × ?	3 Prime factors of 12 — how many, counting repeats	4 8 as 2 × 2 × ?
5 Prime factors of 10 — the larger one	6 Prime factors of 15 — the smaller	7 5 × 7	8 8 × 9
9 6 × 5	10 9 × 4	11 7 × 2	12 8 × 5

Core GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Prime factors of 36 counting repeats (2×2×3×3) ....
14. Largest prime factor of 60 ....
15. Largest prime factor of 84 ....
16. Prime factors of 100 counting repeats ....
17. Largest prime factor of 45 ....
18. 16 × 12 ....
19. 25 × 14 ....
20. 33 × 12 ....
21. 27 × 15 ....
22. 44 × 13 ....

★ Challenge

NOT GRADED · REASON IT OUT

23. Largest prime factor of 210

24. 2 × 2 × 3 × 5 equals

25. 58 × 46

26. 91 × 37

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Twin Primes

Enrichment. Pairs two apart — and nobody knows whether they ever stop.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

4 ITEMS · RECALL

1 The prime two above 3

2 The prime two above 5

3 6 × 3

4 9 × 5

Core

GRADED · SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. The prime two above 11

6. The prime two above 17

7. 26 × 12

Challenge

NOT GRADED · REASON IT OUT

8. Twin prime pairs below 50 — how many
9. The prime two above 29
10. The only prime three apart from another prime (2 and 5)
11. 18 × 9
12. 64 × 25
13. 73 × 42
14. 500 × 6

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 · Multiples

OUT OF 14

Warm-Up · 1–9 · 12 | 15 | 24 | yes | no | 9 | 40 | 24 | 28

Core · 10–14 · 12 | 15 | 24 | 7 | 20

Challenge · 15–26 · 50 | 36 | 8 | 24 | 90 | 91 | 102 | 152 | 198 | 217 | 984 | 1842

4.2 · Divisibility Rules

OUT OF 22

Warm-Up · 1–12 · yes | yes | no | yes | 6 | yes | 8 | 15 | 30 | 30 | 16 | 12

Core · 13–22 · 18 | yes | yes | no | 0 | 84 | 90 | 120 | 162 | 140

Challenge · 23–26 · 6 | yes | 1000 | 564

4.3 · Primes & Composites

OUT OF 16

Warm-Up · 1–6 · yes | no | yes | no | 2 | 2

Core · 7–16 · 8 | 15 | 29 | no | 97 | 132 | 154 | 253 | 385 | 462

Challenge · 17–26 · 25 | no | 22 | 33 | 44 | 55 | 66 | 77 | 957 | 1353

4.4 · Prime Factor Trees

OUT OF 22

Warm-Up · 1–12 · 6 | 3 | 3 | 2 | 5 | 3 | 35 | 72 | 30 | 36 | 14 | 40

Core · 13–22 · 4 | 5 | 7 | 4 | 5 | 192 | 350 | 396 | 405 | 572

Challenge · 23–26 · 7 | 60 | 2668 | 3367

Fri · Twin Primes

OUT OF 8

Warm-Up · 1–4 · 5 | 7 | 18 | 45

Core · 5–8 · 13 | 19 | 162 | 312

Challenge · 9–14 · 6 | 31 | 2 | 1600 | 3066 | 3000

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Missing-Digit Puzzles

Reason backwards from the product. The last digit gives it away.

Warm-Up

10 ITEMS · RECALL

1 4 × ? = 24

2 ? × 7 = 56

3 9 × ? = 81

4 ? × 6 = 42

5 12 × ? = 60

6 ? × 11 = 88

7 4 × 6

8 3 × 8

9 5 × 4

10 9 × 3

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 3 × 4? = 129 — the missing digit ....

12. ? × 14 = 322 — the number ....

13. 6 × 4? = 276 — the missing digit ....

14. 2? × 3 = 84 — the missing digit ....

15. ? × 12 = 156 ....

★ Challenge

NOT GRADED · REASON IT OUT

16. ?? × ?? = 1,024 using two equal 2-digit numbers — type one
17. 4? × 5 ends in 5. The missing digit can be 1, 3, 5, 7 or 9 — type the smallest
18. 7 × 10
19. 2 × 25
20. 21 × 8
21. 34 × 6
22. 47 × 5
23. 56 × 7
24. 38 × 9
25. 216 × 9
26. 408 × 7

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Launch Bay Blueprint

Design the hangar, then find its area two different ways.

Warm-Up

6 ITEMS · RECALL

1 Area of a 10 by 8 bay

2 Area of a 12 by 5 bay

3 Perimeter of a 10 by 8 bay

4 Area of a 6 by 6 bay

5 Area of a 20 by 3 bay

6 Perimeter of a 6 by 6 bay

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

7. A 14 by 12 hangar — area
8. That hangar split at 10: the 10 by 12 part
9. And the 4 by 12 part
10. Do the two parts add to the whole — type the total
11. A 23 by 14 hangar — area
12. 24 × 15
13. 36 × 14
14. 45 × 16
15. 53 × 12
16. 28 × 25

Challenge

NOT GRADED · REASON IT OUT

17. A hangar of area 144 with equal sides — one side
18. Double both sides of a 14 by 12 hangar. Area multiplies by
19. 10×4
20. 20×6
21. 30×5
22. 40×4
23. 50×6
24. 70×3
25. 96×45
26. 84×63

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Blueprint Defence

Both methods must agree, and you have to say why they do.

Warm-Up

9 ITEMS · RECALL

1 20 × 10	2 3 × 4	3 Sum of those four rooms	4 23 × 14 by algorithm
5 5 × 5	6 6 × 7	7 8 × 2	8 4 × 9
9 7 × 7			

Core

GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. 36 × 25 by area model — the total ....
11. 36 × 25 by algorithm ....
12. The tens×tens room of 36 × 25 ....
13. The ones×ones room of 36 × 25 ....
14. 47 × 18 ....
15. 19 × 12 ....
16. 28 × 13 ....
17. 37 × 14 ....
18. 46 × 15 ....
19. 52 × 17 ....

Challenge

NOT GRADED · REASON IT OUT

20. 20×4

21. 3×10

22. Rooms in a 3-digit by 2-digit model

23. 124×23

24. 125×16

25. 144×12

Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal

ONE SENTENCE PER MISTAKE

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up
9 ITEMS · RECALL

1 8 × 7

2 Is 9 prime — yes or no

3 Factors of 16 — how many

4 Third multiple of 7

5 2 × 5

6 4 × 3

7 6 × 8

8 9 × 9

9 7 × 8

Core
GRADED · SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10. 38 × 24
11. GCF of 24 and 36
12. Largest prime factor of 84
13. 23 × 16
14. 41 × 18
15. 35 × 21
16. 29 × 24
17. 48 × 19

★ Challenge

NOT GRADED · REASON IT OUT

18. 6 × 40
19. 5 × 18
20. 9 × 26
21. Estimate 296 × 5
22. Primes below 100 — how many
23. 87 × 64
24. 5 × 10
25. 67 × 54
26. 250 × 24

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Mission 01 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
RIGHT ____ OF ____ TRIED

Warm-Up

3 ITEMS · RECALL

1 7 × 9

2 8 × 7

3 6 × 9

Core

GRADED · SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

4. 23 × 14
5. 45 × 23
6. Factor pairs of 36 — how many
7. GCF of 16 and 40
8. Is 51 prime — yes or no
9. Largest prime factor of 60
10. 32 × 14
11. 27 × 23

★ Challenge

NOT GRADED · REASON IT OUT

12. 8×30
13. 6×47
14. 326×4
15. Area 84, one side 6 — the other
16. Estimate 62×38, then give the true value
17. 88×45
18. 125×32
19. 999×9

✎ Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

! Error journal

ONE SENTENCE PER MISTAKE

Week 5 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

5.1 · Missing-Digit Puzzles

OUT OF 15

Warm-Up · 1–10 · 6 | 8 | 9 | 7 | 5 | 8 | 24 | 24 | 20 | 27

Core · 11–15 · 3 | 23 | 6 | 8 | 13

Challenge · 16–26 · 32 | 1 | 70 | 50 | 168 | 204 | 235 | 392 | 342 | 1944 | 2856

5.2 · Launch Bay Blueprint

OUT OF 16

Warm-Up · 1–6 · 80 | 60 | 36 | 36 | 60 | 24

Core · 7–16 · 168 | 120 | 48 | 168 | 322 | 360 | 504 | 720 | 636 | 700

Challenge · 17–26 · 12 | 4 | 40 | 120 | 150 | 160 | 300 | 210 | 4320 | 5292

5.3 · Blueprint Defence

OUT OF 19

Warm-Up · 1–9 · 200 | 12 | 322 | 322 | 25 | 42 | 16 | 36 | 49

Core · 10–19 · 900 | 900 | 600 | 30 | 846 | 228 | 364 | 518 | 690 | 884

Challenge · 20–25 · 80 | 30 | 6 | 2852 | 2000 | 1728

Thu · Error Journal Sweep

OUT OF 22

Warm-Up · 1–12 · 56 | 240 | no | 5 | 90 | 21 | 10 | 12 | 48 | 81 | 56 | 50

Core · 13–22 · 234 | 912 | 12 | 7 | 1500 | 368 | 738 | 735 | 696 | 912

Challenge · 23–26 · 25 | 5568 | 3618 | 6000

Fri · Mission 01 Test

OUT OF 14

Warm-Up · 1–4 · 63 | 240 | 56 | 54

Core · 5–14 · 282 | 322 | 1035 | 1304 | 5 | 8 | no | 5 | 448 | 621

Challenge · 15–19 · 14 | 2356 | 3960 | 4000 | 8991